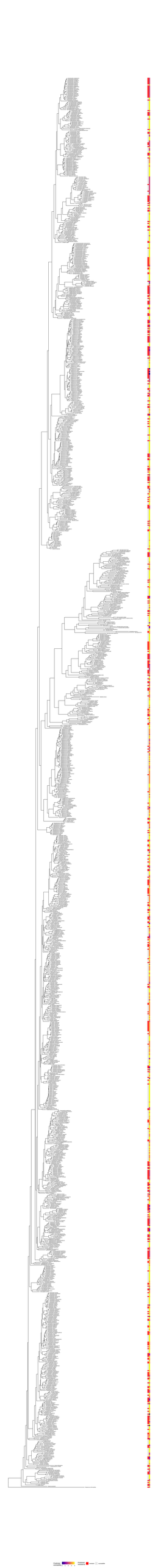

Hydrogenibacillus schlegelii

Brockia lithotrophica
Calditerricola satsumensis

Tepidibacillus fermentans

Vulcanibacillus modesticaldus
Microaerobacter geothermalis

Ammoniphilus resinae

Aneurinibacillus aneurinilyticus

Aneurinibacillus migulanus
Aneurinibacillus danicus
Aneurinibacillus thermoaerophilus
Aneurinibacillus soli
Aneurinibacillus tyrosinisolvens

Brevibacillus humidisoli

Brevibacillus marinus

Brevibacillus ruminantium

Brevibacillus borstelensis

Brevibacillus gelatini

Brevibacillus parabrevis

Brevibacillus fortis

Brevibacillus antibioticus

Brevibacillus porteri

Brevibacillus formosus

Brevibacillus brevis

Brevibacillus reuszeri
Brevibacillus choshinensis

Brevibacillus centrosporus

Brevibacillus panacihumi

Brevibacillus invocatus

Brevibacillus thermoruber
Brevibacillus fulvus
Brevibacillus laterosporus

Brevibacillus dissolubilis
Brevibacillus fluminis

Rubeoparvulum massiliense

Caenibacillus caldisaponilyticus

Scopulibacillus darangshiensis

Scopulibacillus daqui

Sporolactobacillus pectinivorans

Sporolactobacillus shoreae
Sporolactobacillus vineae

Sporolactobacillus shoreicorticis

Sporolactobacillus laevolacticus
Sporolactobacillus mangiferae
Sporolactobacillus terrae

Sporolactobacillus inulinus

Terrilactibacillus tamarindi

Lottiidibacillus patelloidae

Massilibacterium senegalense

Fictibacillus macauensis

Fictibacillus norfolkensis
Bacillus taeanensis

Alteribacter lacisalsi

Alteribacter natronophilus
Alteribacter keqinensis
Alteribacter populi

Evansella cellulosilytica

Evansella clarkii

Evansella halocellulosilytica

Bacillus shivajii

Alkalicoccus urumqiensis

Alkalicoccus luteus
Paenalkalicoccus suaedae
Salisediminibacterium beveridgei

Salisediminibacterium selenitireducens

Salipaludibacillus agaradhaerens

Salipaludibacillus neizhouensis

Texcoconibacillus texcoconensis
Salicibibacter kimchii

Salicibibacter halophilus

Salicibibacter cibi

Salicibibacter cibarius
Geomicrobium halophilum
Geomicrobium sediminis
Salsuginibacillus halophilus

Marinococcus luteus

Marinococcus halophilus

Salibacterium halotolerans

Alteribacillus bidgolensis

Alteribacillus persepolensis

Bacillus piscicola

Alteribacillus iranensis

Halalkalibacter urbisdiaboli

Bacillus solitudinis

Halalkalibacter hemicellulosilyticus

Halalkalibacter akibai

Halalkalibacter alkaliphilus

Halalkalibacter okhensis
Halalkalibacter wakoensis

Alkalihalobacillus alcalophilus

Alkalihalobacillus pseudalcaliphilus
Alkalihalobacillus trypoxylicola
Alkalihalophilus marmarensis

Alkalihalophilus pseudofirmus

Shouchella miscanthi

Shouchella patagoniensis

Shouchella clausii

Shouchella rhizosphaerae
Shouchella shacheensis
Halalkalibacterium ligniniphilum

Halalkalibacterium halodurans

Alkalihalobacterium alkalinitrilicum

Alkalihalobacterium bogoriense

Alkalihalobacterium alkalicellulosilyticum

Desertibacillus haloalkaliphilus
Anaerobacillus isosaccharinicus

Anaerobacillus arseniciselenatis

Bacillus mesophilus

Bacillus pinisoli

Ectobacillus funiculus

Bacillus fungorum

Bacillus tropicus

Bacillus anthracis

Bacillus paranthracis

Bacillus albus

Bacillus dicomae

Bacillus sanguinis

Bacillus proteolyticus

Bacillus paramycoides

Bacillus mycoides

Bacillus hominis

Bacillus toyonensis
Bacillus wiedmannii

Bacillus mobilis
Bacillus luti

Bacillus thuringiensis
Bacillus clarus
Bacillus pseudomycoides

Bacillus bingmayongensis
Bacillus cytotoxicus
Bacillus cereus

Bacillus rhizoplanae
Bacillus manliponensis

Priestia flexa

Priestia megaterium

Priestia aryabhattai

Priestia iocasae
Priestia abyssalis

Sutcliffiella halmapala

Sutcliffiella rhizosphaerae

Sutcliffiella horikoshii

Bacillus alkalisoli
Sutcliffiella cohnii

Aeribacillus composti

Bacillus stratosphericus

Bacillus safensis

Bacillus xiamenensis

Bacillus altitudinis

Bacillus pumilus

Bacillus tequilensis

Bacillus vallismortis
Bacillus cabrialesii
Bacillus spizizenii

Bacillus inaquosorum
Bacillus halotolerans
Bacillus siamensis

Bacillus velezensis
Bacillus amyloliquefaciens
Bacillus nakamurai

Bacillus paralicheniformis

Bacillus haynesii
Bacillus licheniformis
Bacillus glycinifermentans

Bacillus sonorensis

Metabacillus flavus

Metabacillus mangrovi
Metabacillus lacus

Metabacillus dongyingensis

Metabacillus idriensis
Metabacillus indicus

Metabacillus elymi

Metabacillus sediminilitoris

Metabacillus crassostreae

Bacillus weihaiensis

Metabacillus halosaccharovorans

Metabacillus niabensis
Metabacillus schmidteae

Metabacillus fastidiosus

Bacillus timonensis

Litchfieldia sinesaloumensis
Litchfieldia alkalitelluris

Anoxybacillus flavithermus

Anoxybacillus mongoliensis

Anoxybacillus gonensis

Anoxybacillus ayderensis
Anoxybacillus pushchinoensis

Parageobacillus thermoglucosidasius

Parageobacillus toebii

Geobacillus subterraneus

Geobacillus thermodenitrificans

Geobacillus zalihae

Geobacillus thermocatenulatus
Geobacillus kaustophilus
Geobacillus vulcani
Geobacillus jurassicus
Geobacillus stearothermophilus

Saccharococcus thermophilus

Anoxybacillus calidus

Pradoshia eiseniae

Peribacillus saganii

Bacillus massiliglaciei

Peribacillus asahii

Peribacillus simplex

Peribacillus frigoritolerans
Peribacillus butanolivorans

Peribacillus loiseleuriae

Peribacillus huizhouensis
Peribacillus psychrosaccharolyticus
Peribacillus cavernae

Peribacillus glennii

Peribacillus deserti
Bacillus massiliigorillae

Bacillus massilioanorexius
Bacillus testis
Peribacillus alkalitolerans

Peribacillus acanthi

Mesobacillus foraminis

Mesobacillus maritimus

Bacillus mediterraneensis

Mesobacillus zeae

Mesobacillus subterraneus

Mesobacillus jeotgali

Mesobacillus boroniphilus

Mesobacillus selenatarsenatis

Mesobacillus harenae

Bacillus marasmi

Neobacillus notoginsengisoli

Neobacillus terrae

Neobacillus dielmonensis

Neobacillus muris

Neobacillus citreus

Neobacillus rhizophilus

Bacillus rubiinfantis

Neobacillus kokaensis

Neobacillus mesonae

Neobacillus fumarioli

Neobacillus endophyticus

Neobacillus niacini

Neobacillus soli

Neobacillus novalis

Neobacillus rhizosphaerae

Neobacillus drentensis

Neobacillus jeddahensis
Neobacillus bataviensis

Neobacillus massiliamazoniensis

Bacillus methanolicus

Bacillus canaveralius

Bacillus massilionigeriensis

Bacillus tuaregi

Bacillus benzoevorans

Niallia nealsonii

Niallia circulans

Niallia alba

Robertmurraya siralis

Robertmurraya kyonggiensis
Robertmurraya dakarensis
Robertmurraya korlensis

Bacillus infantis

Cytobacillus firmus

Sporosarcina globispora

Cytobacillus depressus

Cytobacillus dafuensis

Cytobacillus citreus

Cytobacillus solani

Cytobacillus praedii
Cytobacillus massiliigabonensis
Cytobacillus eiseniae

Cytobacillus stercorigallinarum

Cytobacillus kochii
Cytobacillus gottheilii

Bacillus aquiflavi

Bacillus litorisediminis

Bacillus oleivorans
Bacillus fonticola

Bacillus andreraoultii

Pallidibacillus pasinlerensis
Bacillus kwashiorkori

Psychrobacillus psychrotolerans

Chryseomicrobium excrementi

Chryseomicrobium aureum

Paenisporosarcina antarctica

Paenisporosarcina cavernae

Lysinibacillus odysseyi

Solibacillus isronensis

Solibacillus faecavium

Solibacillus merdavium

Solibacillus palustris

Bacillus ndiopicus

Metasolibacillus fluoroglycofenilyticus

Caryophanon latum

Caryophanon tenue

Metalysinibacillus jejuensis

Lysinibacillus cavernae

Lysinibacillus capsici
Lysinibacillus fusiformis
Lysinibacillus macroides
Lysinibacillus agricola

Lysinibacillus xylanilyticus

Lysinibacillus parviboronicapiens

Lysinibacillus contaminans
Lysinibacillus sphaericus

Ureibacillus thermophilus

Ureibacillus terrenus
Ureibacillus thermosphaericus

Lysinibacillus telephonicus

Ureibacillus xyleni

Lysinibacillus antri

Ureibacillus chungkukjangi

Ureibacillus yapensis

Lysinibacillus halotolerans

Ureibacillus galli

Lysinibacillus timonensis
Lysinibacillus composti

Viridibacillus arvi

Viridibacillus arenosi

Rummeliibacillus pycnus

Rummeliibacillus stabekisii

Kurthia sibirica

Kurthia gibsonii

Kurthia zopfii

Kurthia huakuii

Kurthia senegalensis

Planococcus koreensis

Planococcus chinensis
Planococcus ruber

Planococcus donghaensis

Planococcus halocryophilus
Planococcus versutus
Planococcus kocurii
Planococcus antarcticus
Planococcus soli
Planococcus wigleyi

Planococcus halotolerans

Planococcus salinarum
Planococcus beigongshangi

Planococcus plakortidis

Planococcus rifietoensis

Planococcus citreus

Sporosarcina psychrophila

Sporosarcina limicola

Sporosarcina quadrami

Sporosarcina luteola
Sporosarcina cyprini

Sporosarcina koreensis

Sporosarcina gallistercoris

Sporosarcina obsidiansis

Sporosarcina cascadiensis

Sporosarcina ureae

Sporosarcina jiandibaonis

Sporosarcina pasteurii

Sporosarcina ureilytica

Savagea serpentis
Bhargavaea cecembensis

Bacillus xiapuensis

Pseudobacillus wudalianchiensis

Bacillus aerolatus
Bacillus thermotolerans
Bacillus ectoiniformans

Domibacillus aminovorans

Domibacillus mangrovi
Domibacillus epiphyticus
Domibacillus antri

Domibacillus enclensis

Domibacillus tundrae

Domibacillus iocasae

Rossellomorea aquimaris

Bacillus haikouensis
Rossellomorea marisflavi
Bacillus salacetis

Rossellomorea vietnamensis
Rossellomorea pakistanensis

Mangrovibacillus cuniculi

Jeotgalibacillus proteolyticus

Jeotgalibacillus campisalis

Jeotgalibacillus terrae

Jeotgalibacillus alimentarius
Jeotgalibacillus salarius

Falsibacillus pallidus

Heyndrickxia coagulans

Heyndrickxia ginsengihumi
Heyndrickxia acidicola
Heyndrickxia vini

Heyndrickxia sporothermodurans
Heyndrickxia oleronia

Lederbergia wuyishanensis

Lederbergia citrisecunda
Lederbergia citri
Lederbergia citrea
Lederbergia galactosidilytica

Bacillus niameyensis
Lederbergia lenta
Bacillus norwichensis

Siminovitchia terrae
Siminovitchia acidinfaciens

Bacillus smithii

Calidifontibacillus erzurumensis

Calidifontibacillus oryziterrae

Bacillus solimangrovi

Bacillus marinisedimentorum

Litoribacterium kuwaitense

Aureibacillus halotolerans

Thalassobacillus cyri

Thalassobacillus pellis

Salimicrobium humidisoli

Halobacillus halophilus

Halobacillus alkaliphilus

Halobacillus aidingensis

Halobacillus salinus

Halobacillus locisalis

Halobacillus mangrovi

Halobacillus yeomjeoni

Halobacillus salinarum

Halobacillus massiliensis

Halobacillus shinanisalinarum

Halobacillus amylolyticus

Halobacillus ihumii

Halobacillus hunanensis

Aquibacillus sediminis

Sediminibacillus dalangtanensis

Sediminibacillus halophilus
Sediminibacillus terrae
Virgibacillus senegalensis
Sediminibacillus massiliensis

Radiobacillus deserti

Radiobacillus kanasensis

Paraliobacillus salinarum

Paraliobacillus zengyii

Amphibacillus cookii

Amphibacillus xylanus

Halolactibacillus halophilus

Halolactibacillus miurensis
Halolactibacillus alkaliphilus
Streptohalobacillus salinus
Pelagirhabdus alkalitolerans

Gracilibacillus salitolerans

Gracilibacillus timonensis
Gracilibacillus halotolerans
Saliterribacillus persicus

Terribacillus halophilus

Pseudogracilibacillus auburnensis

Cerasibacillus quisquiliarum

Cerasibacillus terrae

Lentibacillus saliphilus

Virgibacillus pantothenticus

Virgibacillus dokdonensis
Virgibacillus proomii
Virgibacillus halodenitrificans

Oceanobacillus massiliensis

Oceanobacillus profundus

Oceanobacillus timonensis

Oceanobacillus jeddahense
Oceanobacillus sojae
Oceanobacillus iheyensis

Oceanobacillus kimchii

Oceanobacillus piezotolerans

Oceanobacillus caeni

Oceanobacillus halophilus

Oceanobacillus senegalensis

Oceanobacillus chungangensis

Oceanobacillus zhaokaii
Oceanobacillus alkalisoli
Virgibacillus profundi

Virgibacillus indicus
Virgibacillus salidurans

Lentibacillus jeotgali

Lentibacillus salicampi
Lentibacillus cibarius

Virgibacillus ihumii

Virgibacillus doumboii

Virgibacillus litoralis

Virgibacillus subterraneus

Virgibacillus necropolis

Virgibacillus phasianinus
Virgibacillus alimentarius

Pontibacillus salipaludis

Pontibacillus halophilus

Salirhabdus euzebyi

Salinibacillus kushneri
Salirhabdus salicampi

Tenuibacillus multivorans

Filobacillus milosensis
Halalkalibacillus sediminis
Alkalibacillus haloalkaliphilus

Alkalibacillus almallahensis
Allobacillus halotolerans

Limosilactobacillus pontis

Limosilactobacillus antri

Limosilactobacillus oris

Limosilactobacillus reuteri

Limosilactobacillus balticus
Limosilactobacillus agrestis
Limosilactobacillus caccae

Limosilactobacillus portuensis

Limosilactobacillus urinaemulieris
Limosilactobacillus vaginalis
Limosilactobacillus fastidiosus

Limosilactobacillus avistercoris
Limosilactobacillus frumenti
Limosilactobacillus caecicola

Limosilactobacillus secaliphilus

Limosilactobacillus ingluviei

Limosilactobacillus alvi

Limosilactobacillus equigenerosi

Limosilactobacillus gastricus

Limosilactobacillus gorillae

Limosilactobacillus fermentum

Lactobacillus timonensis

Limosilactobacillus mucosae

Paucilactobacillus nenjiangensis

Paucilactobacillus oligofermentans

Paucilactobacillus hokkaidonensis

Paucilactobacillus wasatchensis
Paucilactobacillus kaifaensis

Fructilactobacillus florum

Fructilactobacillus lindneri

Acetilactobacillus jinshanensis

Apilactobacillus waqarii

Apilactobacillus zhangqiuensis
Apilactobacillus nanyangensis
Apilactobacillus kunkeei
Apilactobacillus bombintestini
Apilactobacillus xinyiensis

Apilactobacillus ozensis

Lentilactobacillus kribbianus

Lentilactobacillus senioris

Lentilactobacillus kefiri

Lentilactobacillus sunkii
Lentilactobacillus buchneri
Lentilactobacillus parabuchneri
Lentilactobacillus rapi
Lentilactobacillus diolivorans
Lentilactobacillus farraginis

Lentilactobacillus hilgardii
Lentilactobacillus parafarraginis
Lentilactobacillus kosonis

Levilactobacillus bambusae

Levilactobacillus fuyuanensis

Levilactobacillus parabrevis
Levilactobacillus hammesii

Levilactobacillus huananensis

Levilactobacillus paucivorans

Levilactobacillus lindianensis

Levilactobacillus lanxiensis

Levilactobacillus wangkuiensis
Levilactobacillus mulengensis
Levilactobacillus cerevisiae
Levilactobacillus angrenensis
Levilactobacillus andaensis

Levilactobacillus koreensis
Levilactobacillus fujinensis
Levilactobacillus tongjiangensis

Levilactobacillus yonginensis
Levilactobacillus brevis
Levilactobacillus tangyuanensis

Levilactobacillus senmaizukei

Levilactobacillus suantsaii

Levilactobacillus zymae

Levilactobacillus acidifarinae

Levilactobacillus namurensis

Levilactobacillus spicheri

Secundilactobacillus odoratitofui

Secundilactobacillus yichangensis

Secundilactobacillus angelensis

Secundilactobacillus silagei

Secundilactobacillus hailunensis
Secundilactobacillus folii
Secundilactobacillus similis
Secundilactobacillus kimchicus
Secundilactobacillus malefermentans

Pediococcus claussenii

Pediococcus ethanolidurans

Pediococcus parvulus
Pediococcus cellicola
Pediococcus damnosus

Pediococcus inopinatus

Lactiplantibacillus modestisalitolerans

Lactiplantibacillus plajomi

Lactiplantibacillus garii

Lactiplantibacillus xiangfangensis

Lactiplantibacillus paraplantarum

Lactiplantibacillus plantarum
Lactiplantibacillus pentosus
Lactiplantibacillus herbarum
Lactiplantibacillus daoliensis

Lactiplantibacillus nangangensis
Lactiplantibacillus fabifermentans

Lactiplantibacillus daowaiensis

Lactiplantibacillus dongliensis

Lactiplantibacillus songbeiensis

Weissella muntiaci

Weissella oryzae

Weissella fangxianensis

Weissella thailandensis
Weissella paramesenteroides
Weissella sagaensis

Weissella hellenica

Weissella viridescens

Weissella ceti

Weissella diestrammenae

Weissella soli

Oenococcus oeni

Oenococcus sicerae
Oenococcus kitaharae

Leuconostoc fallax

Leuconostoc citreum

Leuconostoc palmae

Leuconostoc lactis

Leuconostoc holzapfelii

Leuconostoc gelidum

Leuconostoc inhae
Leuconostoc miyukkimchii
Leuconostoc kimchii

Leuconostoc rapi

Leuconostoc carnosum
Leuconostoc mesenteroides

Leuconostoc suionicum
Leuconostoc litchii
Leuconostoc falkenbergense

Leuconostoc pseudomesenteroides

Fructobacillus durionis

Fructobacillus fructosus

Fructobacillus papyrifericola

Fructobacillus parabroussonetiae
Fructobacillus papyriferae
Fructobacillus broussonetiae

Fructobacillus apis

Fructobacillus tropaeoli

Fructobacillus americanaquae
Fructobacillus pseudoficulneus
Fructobacillus ficulneus
Convivina intestini

Convivina praedatoris

Periweissella fabalis

Periweissella ghanensis
Periweissella fabaria
Periweissella beninensis
Periweissella cryptocerci
Furfurilactobacillus siliginis

Liquorilactobacillus satsumensis

Liquorilactobacillus oeni

Liquorilactobacillus mali

Liquorilactobacillus hordei
Liquorilactobacillus cacaonum

Liquorilactobacillus ghanensis

Liquorilactobacillus nagelii
Liquorilactobacillus vini
Liquorilactobacillus aquaticus

Liquorilactobacillus uvarum
Liquorilactobacillus sucicola
Liquorilactobacillus capillatus

Ligilactobacillus acidipiscis

Ligilactobacillus araffinosus

Ligilactobacillus aviarius

Ligilactobacillus saerimneri

Ligilactobacillus salivarius

Ligilactobacillus agilis

Ligilactobacillus ruminis

Ligilactobacillus apodemi

Dellaglioa algida

Lacticaseibacillus sharpeae

Lacticaseibacillus manihotivorans

Lacticaseibacillus porcinae

Lacticaseibacillus daqingensis

Lacticaseibacillus absianus
Lacticaseibacillus kribbianus
Lacticaseibacillus camelliae

Lacticaseibacillus hegangensis
Lacticaseibacillus jixianensis
Lacticaseibacillus yichunensis

Lacticaseibacillus mingshuiensis

Lacticaseibacillus saniviri

Lacticaseibacillus brantae

Agrilactobacillus composti

Agrilactobacillus yilanensis
Agrilactobacillus fermenti
Schleiferilactobacillus shenzhenensis

Schleiferilactobacillus harbinensis
Schleiferilactobacillus perolens

Lapidilactobacillus achengensis

Lapidilactobacillus wuchangensis

Lapidilactobacillus gannanensis

Lapidilactobacillus concavus

Lapidilactobacillus mulanensis

Lapidilactobacillus dextrinicus

Lapidilactobacillus bayanensis

Companilactobacillus hulinensis

Companilactobacillus jidongensis

Companilactobacillus nodensis
Companilactobacillus insicii

Companilactobacillus tucceti

Companilactobacillus baiquanensis

Companilactobacillus keshanensis

Companilactobacillus kedongensis

Companilactobacillus pabuli

Companilactobacillus farciminis
Companilactobacillus futsaii

Companilactobacillus musae

Companilactobacillus crustorum

Companilactobacillus mindensis

Companilactobacillus kimchiensis
Companilactobacillus nuruki

Companilactobacillus heilongjiangensis

Companilactobacillus huachuanensis
Companilactobacillus nantensis
Companilactobacillus zhachilii

Companilactobacillus alimentarius

Companilactobacillus kimchii

Companilactobacillus paralimentarius

Companilactobacillus ginsenosidimutans

Companilactobacillus versmoldensis

Companilactobacillus zhongbaensis

Companilactobacillus furfuricola

Companilactobacillus metriopterae

Bombilactobacillus bombi

Bombilactobacillus mellis

Bombilactobacillus thymidiniphilus

Bombilactobacillus folatiphilus
Bombilactobacillus apium

Lactobacillus apis

Lactobacillus panisapium

Lactobacillus bombicola

Lactobacillus kullabergensis

Lactobacillus kimbladii

Lactobacillus delbrueckii

Lactobacillus rodentium

Lactobacillus gasseri

Lactobacillus paragasseri

Lactobacillus johnsonii

Lactobacillus taiwanensis
Lactobacillus colini
Lactobacillus iners

Holzapfeliella floricola

Latilactobacillus curvatus

Latilactobacillus sakei
Latilactobacillus fuchuensis
Lactobacillus selangorensis
Loigolactobacillus iwatensis

Loigolactobacillus backii
Loigolactobacillus zhaoyuanensis

Enterococcus nangangensis

Enterococcus timonensis

Enterococcus canis

Enterococcus thailandicus

Enterococcus ratti

Enterococcus hirae

Enterococcus villorum

Enterococcus durans
Enterococcus mundtii
Enterococcus faecium

Enterococcus lactis

Enterococcus phoeniculicola

Enterococcus asini

Enterococcus dispar

Enterococcus saigonensis

Enterococcus songbeiensis

Enterococcus mediterraneensis

Enterococcus columbae

Enterococcus cecorum

Enterococcus sulfureus

Enterococcus italicus

Enterococcus saccharolyticus

Enterococcus aquimarinus

Enterococcus lemanii

Tetragenococcus halophilus

Tetragenococcus osmophilus
Tetragenococcus koreensis

Enterococcus casseliflavus

Enterococcus gallinarum

Enterococcus alishanensis

Enterococcus malodoratus

Enterococcus hulanensis

Enterococcus raffinosus

Enterococcus avium

Enterococcus gilvus

Enterococcus pingfangensis

Enterococcus xiangfangensis
Enterococcus viikkiensis
Enterococcus hermanniensis

Enterococcus pallens

Enterococcus florum

Enterococcus ureilyticus

Enterococcus silesiacus

Enterococcus crotali

Enterococcus rotai

Enterococcus caccae
Enterococcus moraviensis
Enterococcus ureasiticus

Enterococcus haemoperoxidus
Enterococcus plantarum
Enterococcus quebecensis
Enterococcus wangshanyuanii
Enterococcus rivorum
Enterococcus faecalis
Enterococcus larvae
Melissococcus plutonius

Vagococcus entomophilus

Vagococcus hydrophili

Vagococcus carniphilus
Vagococcus fluvialis
Vagococcus martis

Vagococcus teuberi

Vagococcus penaei

Vagococcus vulneris
Vagococcus silagei
Vagococcus acidifermentans
Vagococcus zengguangii

Vagococcus xieshaowenii
Vagococcus elongatus
Vagococcus humatus
Vagococcus coleopterorum

Vagococcus fessus
Vagococcus salmoninarum

Floricoccus penangensis

Floricoccus tropicus

Lactococcus fujiensis

Lactococcus hircilactis
Lactococcus nasutitermitis

Lactococcus lactis

Lactococcus cremoris
Lactococcus taiwanensis
Lactococcus allomyrinae

Lactococcus protaetiae

Lactococcus petauri

Lactococcus formosensis
Lactococcus ileimucosae
Lactococcus garvieae
Lactovum miscens

Streptococcus parasuis

Streptococcus suis
Streptococcus oriscaviae

Streptococcus acidominimus

Streptococcus marmotae

Streptococcus zhangguiae
Streptococcus respiraculi
Streptococcus cuniculi

Streptococcus gallinaceus

Streptococcus minor

Streptococcus ovis

Streptococcus himalayensis

Streptococcus massiliensis

Streptococcus oricebi

Streptococcus sinensis

Streptococcus sanguinis
Streptococcus panodentis
Streptococcus gordonii

Streptococcus anginosus

Streptococcus constellatus

Streptococcus intermedius

Streptococcus shenyangsis

Streptococcus gwangjuensis

Streptococcus vulneris

Streptococcus pseudopneumoniae

Streptococcus halitosis

Streptococcus cristatus
Streptococcus oralis

Streptococcus lactarius

Streptococcus parasanguinis

Streptococcus caecimuris

Streptococcus ilei

Streptococcus xiaochunlingii
Streptococcus australis
Streptococcus rubneri

Streptococcus orisasini

Streptococcus ratti
Streptococcus mutans
Streptococcus macacae
Streptococcus ferus

Streptococcus marimammalium

Streptococcus pacificus
Streptococcus cuniculipharyngis

Streptococcus hillyeri

Streptococcus porci

Streptococcus salivarius

Streptococcus vestibularis

Streptococcus loxodontisalivarius

Streptococcus saliviloxodontae

Streptococcus pluranimalium

Streptococcus hyovaginalis

Streptococcus sobrinus

Streptococcus downei
Streptococcus criceti

Streptococcus caballi

Streptococcus lutetiensis

Streptococcus infantarius
Streptococcus vicugnae
Streptococcus equinus
Streptococcus macedonicus

Streptococcus gallolyticus
Streptococcus pasteurianus

Streptococcus henryi

Streptococcus orisratti

Streptococcus pantholopis

Streptococcus chenjunshii

Streptococcus agalactiae

Streptococcus hyointestinalis

Streptococcus sciuri

Streptococcus catagoni

Streptococcus didelphis

Streptococcus porcinus

Streptococcus pseudoporcinus

Streptococcus penaeicida

Streptococcus bovimastitidis
Streptococcus uberis

Streptococcus parauberis

Streptococcus iniae

Streptococcus ictaluri

Streptococcus dysgalactiae

Streptococcus pyogenes
Streptococcus canis
Streptococcus castoreus

Streptococcus phocae

Streptococcus equi
Streptococcus halichoeri
Streptococcus urinalis

Catellicoccus marimammalium

Jeotgalibaca porci

Jeotgalibaca dankookensis

Jeotgalibaca arthritidis

Jeotgalibaca ciconiae
Trichococcus patagoniensis

Abiotrophia defectiva

Vaginisenegalia massiliensis

Globicatella sanguinis

Globicatella sulfidifaciens
Fundicoccus ignavus

Facklamia miroungae

Facklamia lactis
Facklamia languida

Eremococcus coleocola

Falseniella ignava

Suicoccus acidiformans

Ignavigranum ruoffiae

Aerococcus viridans

Aerococcus mictus

Aerococcus christensenii

Bavariicoccus seileri

Granulicatella balaenopterae

Granulicatella seriolae

Granulicatella adiacens

Granulicatella elegans

Carnobacterium gallinarum

Carnobacterium maltaromaticum
Carnobacterium divergens

Carnobacterium mobile

Carnobacterium iners

Carnobacterium funditum

Carnobacterium pleistocenium

Carnobacterium alterfunditum
Carnobacterium jeotgali
Carnobacterium viridans
Carnobacterium inhibens

Allofustis seminis

Dolosigranulum pigrum

Alloiococcus otitis

Alkalibacterium pelagium

Alkalibacterium thalassium
Alkalibacterium olivapovliticus

Alkalibacterium kapii

Alkalibacterium putridalgicola

Alkalibacterium subtropicum

Marinilactibacillus kalidii

Marinilactibacillus psychrotolerans

Marinilactibacillus piezotolerans

Listeria ivanovii

Listeria seeligeri

Listeria monocytogenes

Listeria swaminathanii

Listeria marthii
Listeria cossartiae
Listeria welshimeri
Listeria innocua

Listeria farberi

Listeria grayi

Listeria ilorinensis

Listeria kieliensis

Listeria valentina
Listeria aquatica
Listeria floridensis
Listeria fleischmannii

Listeria cornellensis

Listeria riparia

Listeria booriae

Nosocomiicoccus massiliensis

Nosocomiicoccus ampullae

Phocicoccus schoeneichii

Phocicoccus pinnipedialis

Jeotgalicoccus meleagridis

Jeotgalicoccus marinus

Jeotgalicoccus coquinae

Jeotgalicoccus saudimassiliensis
Jeotgalicoccus psychrophilus
Jeotgalicoccus halotolerans
Jeotgalicoccus aerolatus

Salinicoccus carnicancri

Salinicoccus halodurans

Salinicoccus cyprini

Salinicoccus hispanicus

Salinicoccus roseus

Salinicoccus luteus

Salinicoccus halitifaciens

Salinicoccus kekensis
Lacicoccus qingdaonensis
Abyssicoccus albus

Staphylococcus delphini

Staphylococcus pseudintermedius
Staphylococcus intermedius
Staphylococcus cornubiensis
Staphylococcus lutrae
Staphylococcus schleiferi
Staphylococcus rostri

Staphylococcus americanisciuri
Staphylococcus microti
Staphylococcus muscae
Staphylococcus felis

Staphylococcus canis

Staphylococcus ratti

Staphylococcus chromogenes

Staphylococcus hyicus

Staphylococcus agnetis
Staphylococcus massiliensis

Staphylococcus pettenkoferi

Staphylococcus argensis
Staphylococcus auricularis

Staphylococcus ureilyticus

Staphylococcus cohnii
Staphylococcus nepalensis
Staphylococcus equorum

Staphylococcus saprophyticus

Staphylococcus edaphicus
Staphylococcus caeli
Staphylococcus xylosus

Staphylococcus shinii
Staphylococcus pseudoxylosus

Staphylococcus succinus
Staphylococcus gallinarum
Staphylococcus lloydii

Staphylococcus kloosii
Staphylococcus durrellii
Staphylococcus arlettae

Staphylococcus haemolyticus

Staphylococcus borealis
Staphylococcus devriesei
Staphylococcus pragensis

Staphylococcus croceilyticus
Staphylococcus petrasii
Staphylococcus hominis
Staphylococcus lugdunensis
Staphylococcus epidermidis

Staphylococcus saccharolyticus

Staphylococcus capitis

Staphylococcus caprae

Staphylococcus warneri

Staphylococcus pasteuri

Staphylococcus argenteus

Staphylococcus schweitzeri
Staphylococcus simiae

Staphylococcus simulans

Staphylococcus carnosus

Staphylococcus condimenti

Staphylococcus debuckii

Staphylococcus piscifermentans

Mammaliicoccus vitulinus

Mammaliicoccus sciuri
Mammaliicoccus stepanovicii
Mammaliicoccus lentus
Macrococcus epidermidis

Macrococcoides goetzii
Macrococcoides bohemicum
Macrococcoides caseolyticum

Macrococcoides canis
Macrococcus armenti

Macrococcus carouselicus

Macrococcus bovicus

Macrococcus brunensis

Macrococcus equipercicus

Macrococcus lamae

Gemella haemolysans

Gemella morbillorum
Gemella sanguinis
Gemella cuniculi
Gemella massiliensis

Culicoidibacter larvae

Haloplasma contractile

Turicibacter bilis

Turicibacter sanguinis

Paracholeplasma morum

Acholeplasma brassicae

Haploplasma axanthum

Mariniplasma anaerobium

Acholeplasma laidlawii

Acholeplasma granularum
Acholeplasma oculi
Acholeplasma hippikon
Acholeplasma palmae

Erysipelothrix larvae

Erysipelothrix anatis

Erysipelothrix aquatica
Erysipelothrix inopinata
Erysipelothrix amsterdamensis

Erysipelothrix piscisicarius
Erysipelothrix urinaevulpis

Anaerorhabdus furcosa

Bulleidia extructa

Holdemania filiformis

Holdemania massiliensis

Traorella massiliensis

Dielma fastidiosa

Allobaculum stercoricanis

Allobaculum mucilyticum
Ileibacterium valens
Faecalibaculum rodentium
Dubosiella newyorkensis

Floccifex porci

Holdemanella biformis

Holdemanella porci

Faecalicoccus acidiformans

Faecalicoccus pleomorphus

Absicoccus porci

Erysipelothrix rhusiopathiae

Merdibacter massiliensis
Longicatena caecimuris

Amedibacterium intestinale
Amedibacillus dolichus
Copranaerobaculum intestinale

Tannockella kyphosi

Faecalibacillus intestinalis

Faecalibacillus faecis
Thomasclavelia spiroformis
Thomasclavelia saccharogumia

Thomasclavelia cocleata
Thomasclavelia ramosa

Eggerthia catenaformis

Intestinibaculum porci
Beduini massiliensis

Spiroplasma phoeniceum

Spiroplasma chrysopicola

Spiroplasma syrphidicola

Spiroplasma mirum

Spiroplasma eriocheiris

Spiroplasma monobiae

Spiroplasma floricola
Spiroplasma diminutum
Spiroplasma cantharicola
Spiroplasma taiwanense

Spiroplasma clarkii

Spiroplasma tabanidicola

Spiroplasma gladiatoris
Spiroplasma helicoides
Spiroplasma chinense

Spiroplasma culicicola

Spiroplasma apis

Spiroplasma corruscae

Spiroplasma litorale
Spiroplasma turonicum
Spiroplasma alleghenense

Spiroplasma sabaudiense

Mesoplasma syrphidae

Mesoplasma tabanidae

Mesoplasma grammopterae

Mesoplasma florum

Mesoplasma entomophilum

Mesoplasma coleopterae
Mesoplasma melaleucae
Mesoplasma chauliocola
Entomoplasma ellychniae

Mycoplasma cottewii

Mycoplasma yeatsii
Mycoplasma putrefaciens

Mycoplasma leachii

Mycoplasma capricolum
Mycoplasma mycoides
Mycoplasma feriruminatoris

Entomoplasma freundtii

Williamsoniiplasma lucivorax

Williamsoniiplasma luminosum

Williamsoniiplasma somnilux
Mesoplasma lactucae

Spiroplasma endosymbiont of Nebria riversi
Spiroplasma platyhelix

Spiroplasma attinicola
Spiroplasma endosymbiont of Danaus chrysippus

Malacoplasma penetrans

Malacoplasma iowae

Ureaplasma urealyticum

Ureaplasma parvum
Ureaplasma diversum

Mycoplasmoides pirum

Mycoplasmoides genitalium

Mycoplasmoides pneumoniae

Mycoplasma tullyi

Mycoplasmoides gallisepticum

Mycoplasma ovis

Mycoplasma parvum
Mycoplasmoides fastidiosum

Mycoplasma todarodis

Mycoplasma mobile

Mycoplasmopsis citelli

Mycoplasmopsis columboralis
Mycoplasmopsis cricetuli
Mycoplasmopsis sturni
Mycoplasmopsis synoviae

Mycoplasmopsis gallopavonis

Mycoplasmopsis glycophila

Mycoplasmopsis verecunda

Mycoplasma nasistruthionis

Mycoplasmopsis felis

Mycoplasmopsis mustelae

Mycoplasmopsis cynos

Mycoplasmopsis canis
Mycoplasmopsis bovirhinis
Mycoplasma leonicaptivi

Mycoplasmopsis gallinacea

Mycoplasmopsis pullorum

Mycoplasma anserisalpingitidis

Mycoplasmopsis anatis

Mycoplasma crocodyli

Mycoplasmopsis columbina

Mycoplasmopsis columbinasalis

Mycoplasmopsis meleagridis

Mycoplasmopsis iners
Mycoplasmopsis gallinarum

Mycoplasmopsis lipofaciens

Mycoplasmopsis fermentans

Mycoplasmopsis caviae

Mycoplasmopsis agalactiae

Mycoplasmopsis bovis
Mycoplasmopsis primatum
Mycoplasma tauri
Mycoplasmopsis maculosa

Mycoplasmopsis mucosicanis

Mycoplasmopsis phocirhinis

Mycoplasma miroungigenitalium

Mycoplasma zalophidermidis

Mycoplasmopsis bovigenitalium

Mycoplasmopsis californica

Mycoplasma simbae

Mycoplasmopsis opalescens

Mycoplasmopsis equigenitalium

Mycoplasma elephantis

Mycoplasmopsis agassizii

Mycoplasmopsis pulmonis

Metamycoplasma subdolum

Mycoplasma falconis

Metamycoplasma spumans

Metamycoplasma auris

Metamycoplasma alkalescens

Metamycoplasma canadense

Mycoplasmopsis arginini

Mycoplasma enhydrae

Metamycoplasma phocicerebrale
Mycoplasma struthionis
Mycoplasma phocoeninasale

Mycoplasma phocae

Metamycoplasma hominis
Metamycoplasma arthritidis
Metamycoplasma cloacale

Mycoplasma anseris

Metamycoplasma salivarium

Metamycoplasma hyosynoviae
Metamycoplasma orale
Mycoplasma miroungirhinis

Mycoplasma zalophi
Mycoplasma phocoenae

Mesomycoplasma molare

Mycoplasma collis

Mesomycoplasma neurolyticum

Mycoplasma iguanae

Mesomycoplasma bovoculi

Mesomycoplasma hyopneumoniae

Mesomycoplasma dispar

Exiguobacterium artemiae

Exiguobacterium antarcticum
Exiguobacterium undae
Exiguobacterium oxidotolerans
Exiguobacterium indicum

Exiguobacterium acetylicum
Exiguobacterium flavidum

Exiguobacterium chiriqhucha

Exiguobacterium aurantiacum

Exiguobacterium alkaliphilum

Exiguobacterium aestuarii

Exiguobacterium marinum
Exiguobacterium algae

Caldalkalibacillus mannanilyticus

Caldalkalibacillus salinus
Caldalkalibacillus thermarum

Insulibacter thermoxylanivorax

Xylanibacillus composti

Chengkuizengella marina

Chengkuizengella sediminis
Longirhabdus pacifica

Gorillibacterium timonense

Gorillibacterium massiliense
Paenibacillus senegalensis
Paenibacillus ginsengarvi

Paenibacillus flagellatus
Paenibacillus koleovorans

Paenibacillus larvae

Paenibacillus chitinolyticus

Paenibacillus lutrae

Paenibacillus radicibacter

Paenibacillus albiflavus

Paenibacillus alginolyticus

Paenibacillus phytorum

Paenibacillus whitsoniae

Paenibacillus plantarum

Paenibacillus planticolens

Paenibacillus chondroitinus

Paenibacillus hexagrammi

Paenibacillus solanacearum

Paenibacillus allorhizosphaerae
Paenibacillus cremeus

Paenibacillus naphthalenovorans

Paenibacillus oleatilyticus

Paenibacillus elgii
Paenibacillus ehimensis

Paenibacillus mucilaginosus

Paenibacillus validus

Paenibacillus foliorum

Paenibacillus periandrae

Paenibacillus doosanensis

Paenibacillus rigui

Paenibacillus hamazuiensis

Paenibacillus thalictri

Paenibacillus swuensis

Saccharibacillus alkalitolerans

Saccharibacillus deserti

Saccharibacillus qingshengii
Saccharibacillus brassicae

Paenibacillus bovis

Paenibacillus hunanensis
Paenibacillus nuruki

Paenibacillus sediminis

Paenibacillus lutimineralis

Paenibacillus faecis

Paenibacillus brevis

Paenibacillus antibioticophila

Paenibacillus rubinfantis

Paenibacillus phocaensis
Paenibacillus barengoltzii
Paenibacillus senegalimassiliensis
Paenibacillus oralis

Paenibacillus macerans

Paenibacillus bouchesdurhonensis

Paenibacillus lentus

Paenibacillus ihumii

Paenibacillus woosongensis
Paenibacillus anaericanus

Paenibacillus physcomitrellae

Paenibacillus yonginensis

Paenibacillus zeisoli

Paenibacillus tuaregi
Paenibacillus shirakamiensis

Paenibacillus gallinarum

Paenibacillus illinoisensis

Paenibacillus xylanilyticus

Paenibacillus xylanexedens

Paenibacillus barcinonensis

Paenibacillus kribbensis

Paenibacillus brasilensis
Paenibacillus polymyxa
Paenibacillus terrae

Paenibacillus lemnae

Paenibacillus algicola

Paenibacillus dakarensis

Paenibacillus faecalis
Paenibacillus campinasensis

Paenibacillus antarcticus

Paenibacillus macquariensis
Paenibacillus glacialis
Paenibacillus crassostreae
Paenibacillus dokdonensis

Paenibacillus dendrobii
Paenibacillus azoreducens
Paenibacillus cellulositrophicus

Paenibacillus donghaensis

Paenibacillus monticola
Paenibacillus wynnii
Paenibacillus odorifer

Paenibacillus tritici

Paenibacillus phytohabitans

Paenibacillus borealis

Paenibacillus riograndensis

Paenibacillus jilunlii
Paenibacillus graminis
Paenibacillus camerounensis

Paenibacillus ihuae

Paenibacillus sabinae

Paenibacillus rhizophilus

Paenibacillus durus

Paenibacillus sophorae
Paenibacillus stellifer

Paenibacillus apiarius

Paenibacillus popilliae

Paenibacillus dendritiformis
Paenibacillus thiaminolyticus

Paenibacillus aquistagni

Paenibacillus agilis
Paenibacillus guangzhouensis

Paenibacillus montanisoli

Paenibacillus albus

Paenibacillus glycinis

Paenibacillus lycopersici

Paenibacillus rhizovicinus

Paenibacillus artemisiicola

Paenibacillus oenotherae

Paenibacillus plantiphilus

Paenibacillus methanolicus

Paenibacillus spongiae

Paenibacillus mendelii

Paenibacillus herberti

Paenibacillus pasadenensis

Paenibacillus albicereus

Paenibacillus beijingensis

Paenibacillus arenilitoris

Paenibacillus luteus

Paenibacillus castaneae

Paenibacillus pinisoli

Paenibacillus agaridevorans

Paenibacillus gorillae

Paenibacillus catalpae
Paenibacillus protaetiae

Paenibacillus cisolokensis

Paenibacillus xylaniclasticus

Paenibacillus kobensis
Paenibacillus cellulosilyticus
Paenibacillus curdlanolyticus

Paenibacillus roseus

Paenibacillus darwinianus

Cohnella candidum

Cohnella terricola

Cohnella mopanensis

Cohnella herbarum
Cohnella abietis

Cohnella lubricantis

Cohnella zeiphila
Cohnella nanjingensis
Cohnella algarum
Thermobacillus xylanilyticus

Shimazuella alba

Risungbinella massiliensis

Polycladospora coralii

Staphylospora marina

Laceyella sacchari

Laceyella sediminis
Thermoactinomyces daqus
Thermoactinomyces vulgaris

Thermoactinomyces mirandus

Lihuaxuella thermophila

Paenactinomyces guangxiensis

Thermoflavimicrobium dichotomicum

Seinonella peptonophila

Marinithermofilum abyssi

Paludifilum halophilum

Kroppenstedtia pulmonis

Desmospora activa

Marininema mesophilum

Marininema halotolerans

Melghirimyces algeriensis

Melghirimyces profundicolus
Kroppenstedtia eburnea

Planifilum fimeticola

Polycladomyces abyssicola

Novibacillus thermophilus

Numidum massiliense

Desulfuribacillus stibiiarsenatis

Desulfuribacillus alkaliarsenatis

Kyrpidia tusciae

Kyrpidia spormannii

Effusibacillus dendaii

Tumebacillus avium

Tumebacillus algifaecis

Alicyclobacillus mengziensis

Alicyclobacillus ferrooxydans

Alicyclobacillus cellulosilyticus

Alicyclobacillus macrosporangiidus

Alicyclobacillus kakegawensis

Alicyclobacillus shizuokensis

Alicyclobacillus acidoterrestris

Alicyclobacillus acidocaldarius

Alicyclobacillus sendaiensis
Alicyclobacillus vulcanalis
Alicyclobacillus sacchari
Alicyclobacillus montanus

Pasteuria penetrans
Streptococcus thermophilus

Predicted

evolvability 0 1 2 3 4

Predicted

resistance resistant susceptible